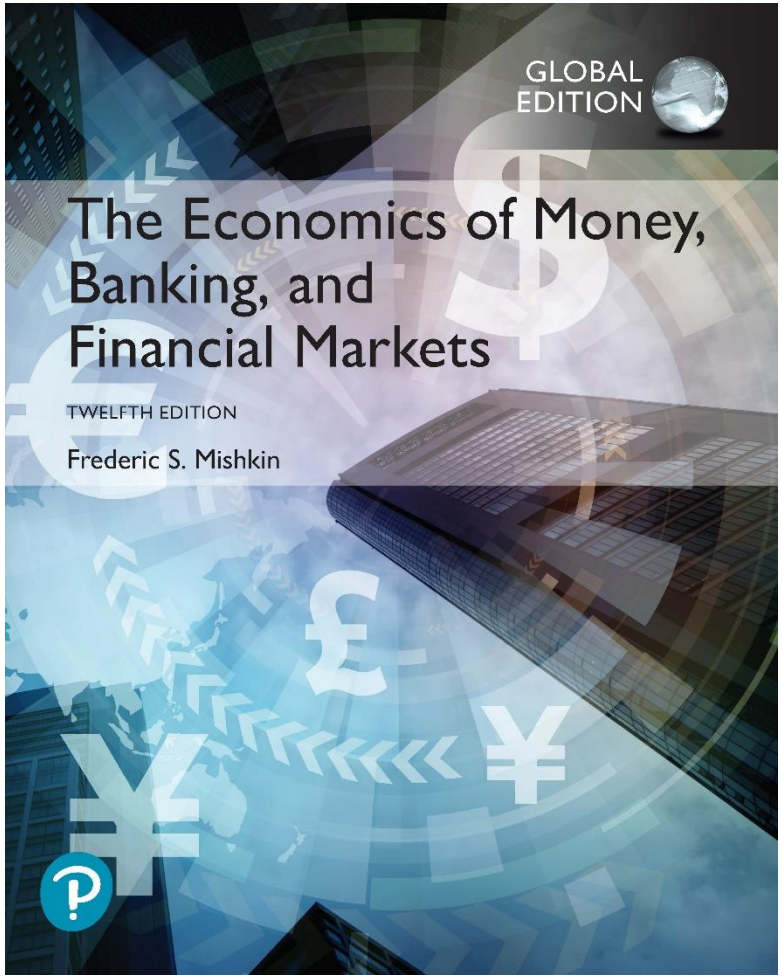
The Economics of Money, Banking, and Financial Markets



Chapter 1

Why Study Money, Banking, and Financial Markets?

Preview

- To examine how the workings of financial markets such as bond, stock and foreign exchange markets affect your everyday life
- To examine how financial institutions such as banks, investment and insurance companies work
- To examine the role of money in the economy

Learning Objectives (1 of 2)

- Recognize the importance of financial markets in the economy.
- Describe how financial intermediation and financial innovation affect banking and the economy.
- Identify the basic links among monetary policy, the business cycle, and economic variables.

Learning Objectives (2 of 2)

- Explain the importance of exchange rates in a global economy.
- Explain how the study of money, banking, and financial markets may advance your career.

Why Study Financial Markets?

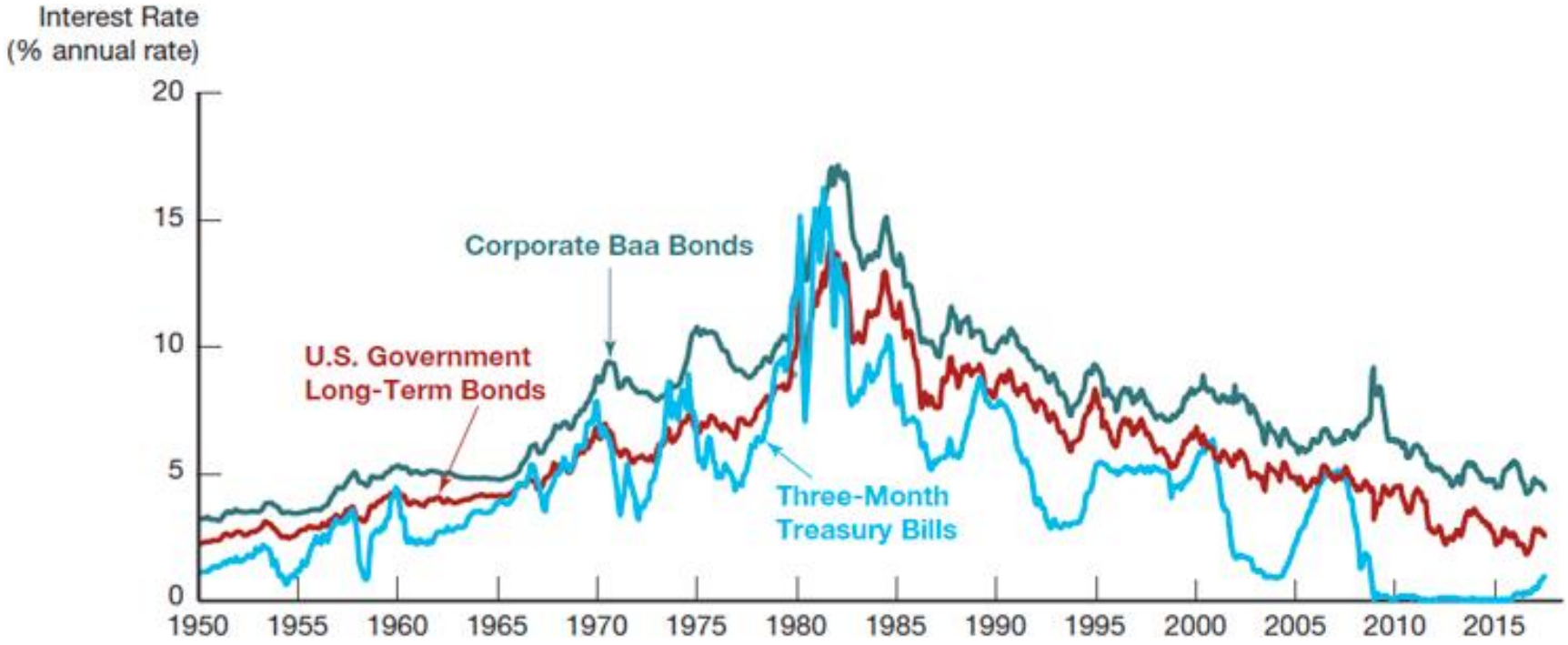
- Financial markets are markets in which funds are transferred from people and firms who have an excess of available funds to people and firms who have a need of funds

--- transferring or exchanging goods/values *across time*

The Bond Market and Interest Rates

- A **security** (financial instrument) is a claim on the issuer's future income or assets.
- A **bond** is a debt security that promises to make payments periodically for a specified period of time.
- An **interest rate** is the cost of borrowing or the price paid for the *rental* of funds.

Figure 1 Interest Rates on Selected Bonds, 1950–2017

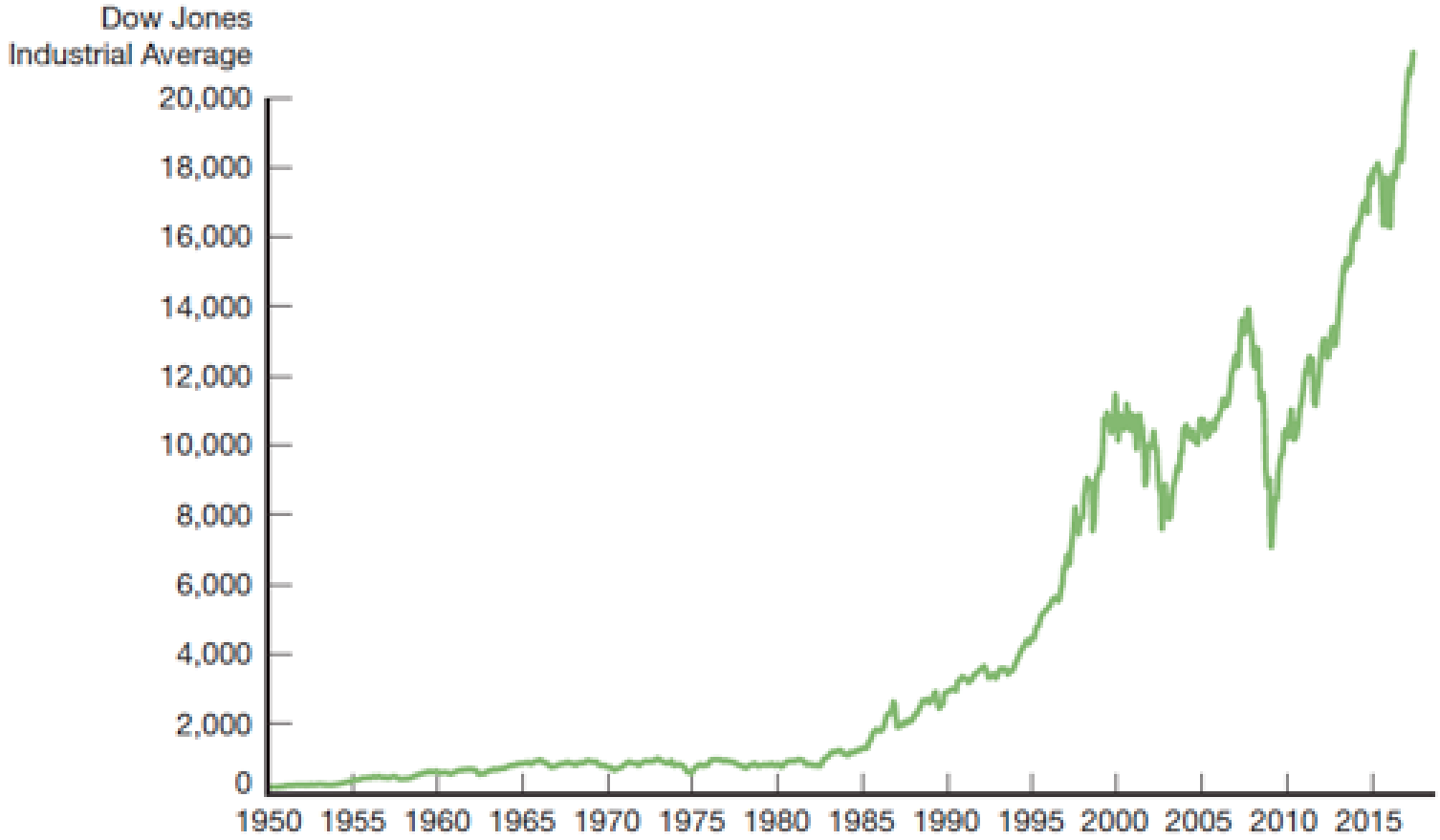


Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/TB3MS; https://fred.stlouisfed.org/series/GS10; https://fred.stlouisfed.org/series/BAA

The Stock Market

- **Common stock** represents a share of ownership in a corporation.
- A **share of stock** is a claim on the residual earnings and assets of the corporation.

Figure 2 Stock Prices as Measured by the Dow Jones Industrial Average, 1950–2017



Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/DJIA

Why Study Financial Institutions and Banking? (1 of 3)

- **Financial intermediaries**: institutions that borrow funds from people who have saved and in turn make loans to people who need funds.
  - **Banks**: accept deposits and make loans
  - Other financial institutions: insurance companies, finance companies, pension funds, mutual funds and investment companies

--- the bridge between savers and fund users

Why Study Financial Institutions and Banking? (2 of 3)

- **Financial innovation**: the development of new financial products and services
  - Can be an important force for good by making the financial system more efficient
  - E-finance: the ability to deliver financial services electronically (e.g., ebanking, WeChat Pay, AliPay)

Good innovation, e.g., convertible debt for high-tech firms

Bad innovation, e.g., some financial products in China's 2015 stock market crash

structured fund (分级基金)

shadow margin financing (场外配资): three layers

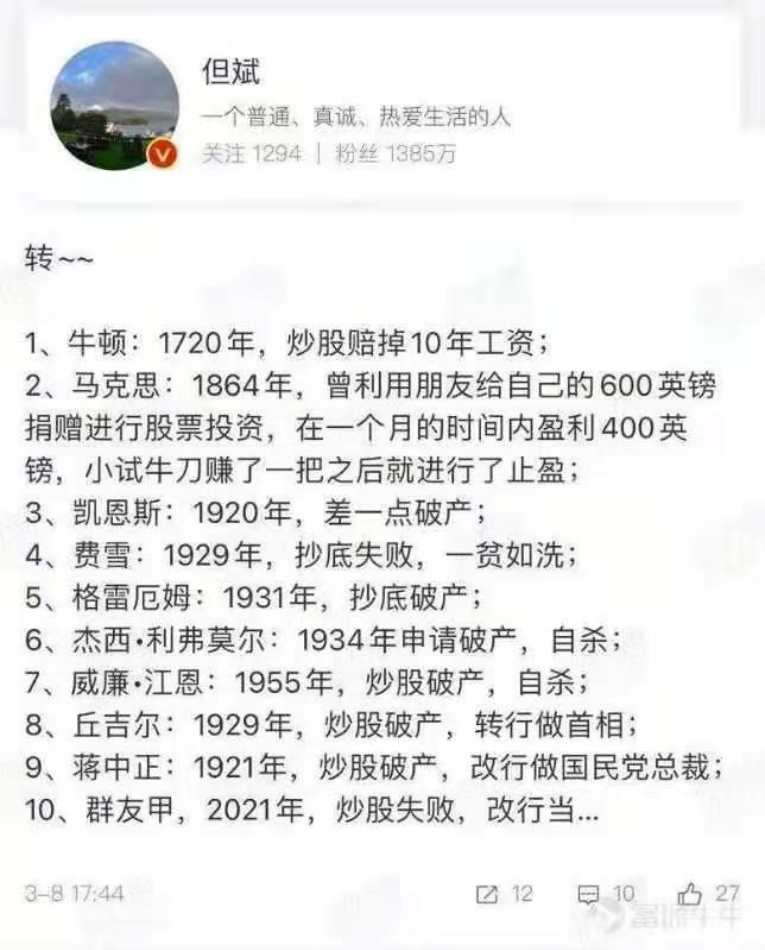
Why Study Financial Institutions and Banking? (3 of 3)

- **Financial crises**: major disruptions in financial markets that are characterized by sharp declines in asset prices and the failures of many financial and nonfinancial firms.

Abound a financial crisis, market sentiment can be too optimistic or too pessimistic, so a huge investment gain/loss can happen.

Famous examples of investment losses in financial crises and market crashes

Entertainment gossips, nevertheless, reflect some truth:



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1、牛顿：1720年，炒股赔掉10年工资；
2、马克思：1864年，曾利用朋友给自己的600英镑捐赠进行股票投资，在一个月的时间内盈利400英镑，小试牛刀赚了一把之后就进行了止盈；
3、凯恩斯：1920年，差一点破产；
4、费雪：1929年，抄底失败，一贫如洗；
5、格雷厄姆：1931年，抄底破产；
6、杰西·利弗莫尔：1934年申请破产，自杀；
7、威廉·江恩：1955年，炒股破产，自杀；
8、丘吉尔：1929年，炒股破产，转行做首相；
9、蒋中正：1921年，炒股破产，改行做国民党总裁；
10、群友甲，2021年，炒股失败，改行当...

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Why Study Money and Monetary Policy?

- Evidence suggests that money, defined as anything that is generally accepted as payment for goods or services or in the repayment of debts, plays an important role in generating business cycles.
- Recessions (unemployment) and expansions affect all of us.
- Monetary theory ties changes in the money supply to changes in aggregate economic activity and the price level.

Stock market performance determinants:

fundamentals (基本面）

liquidity (资金面）

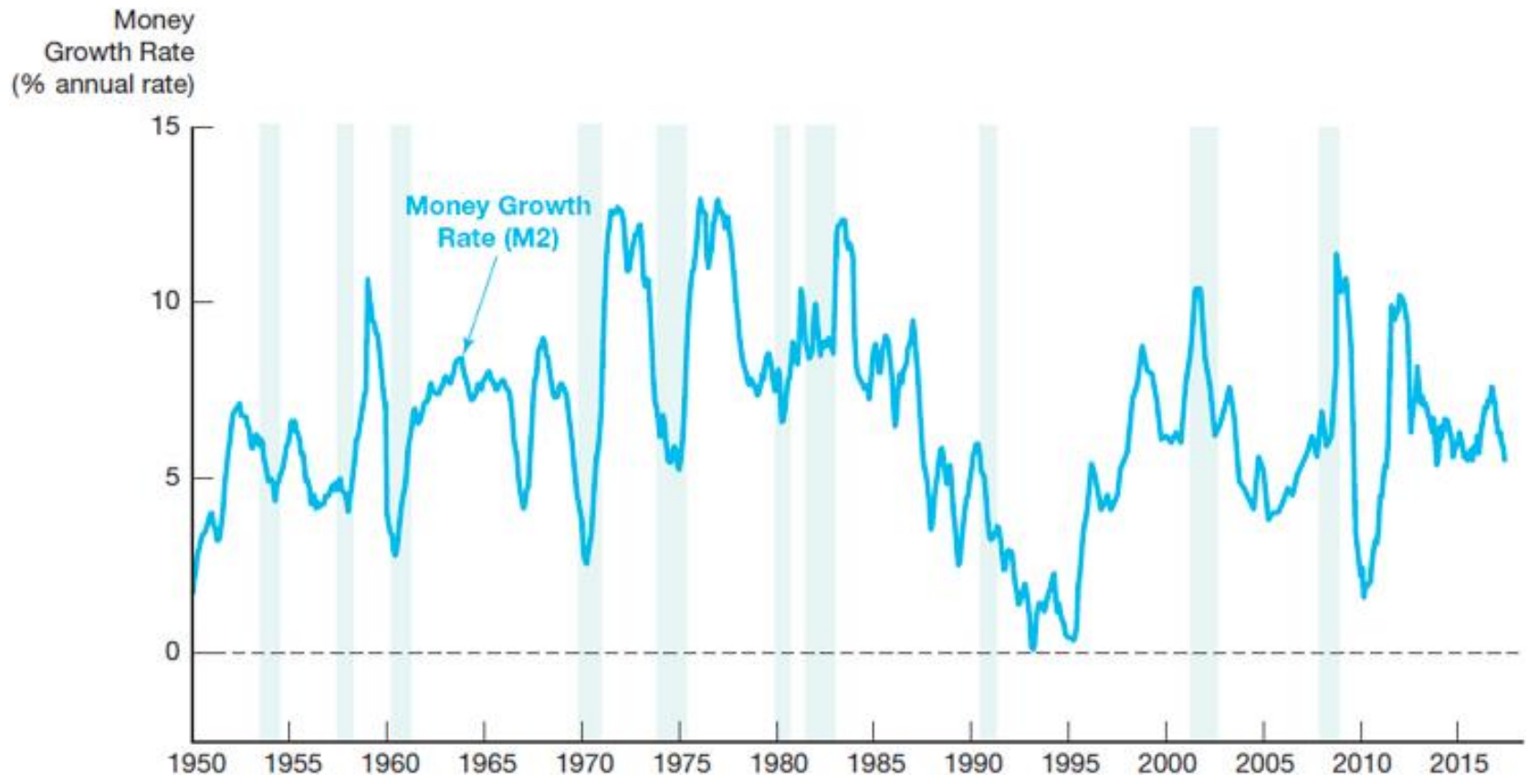
In China, some important measures are M2, new loans (新增贷款), Aggregate Financing to the Real Economy (AFRE) (社会融资规模)
http://www.pbc.gov.cn/diaochatongjisi/116219/116319/index.html

Money, Business Cycles, and Inflation

- The aggregate price level is the average price of goods and services in an economy
- A continual rise in the price level (inflation) affects all economic players
- Data show a connection between the money supply and the price level

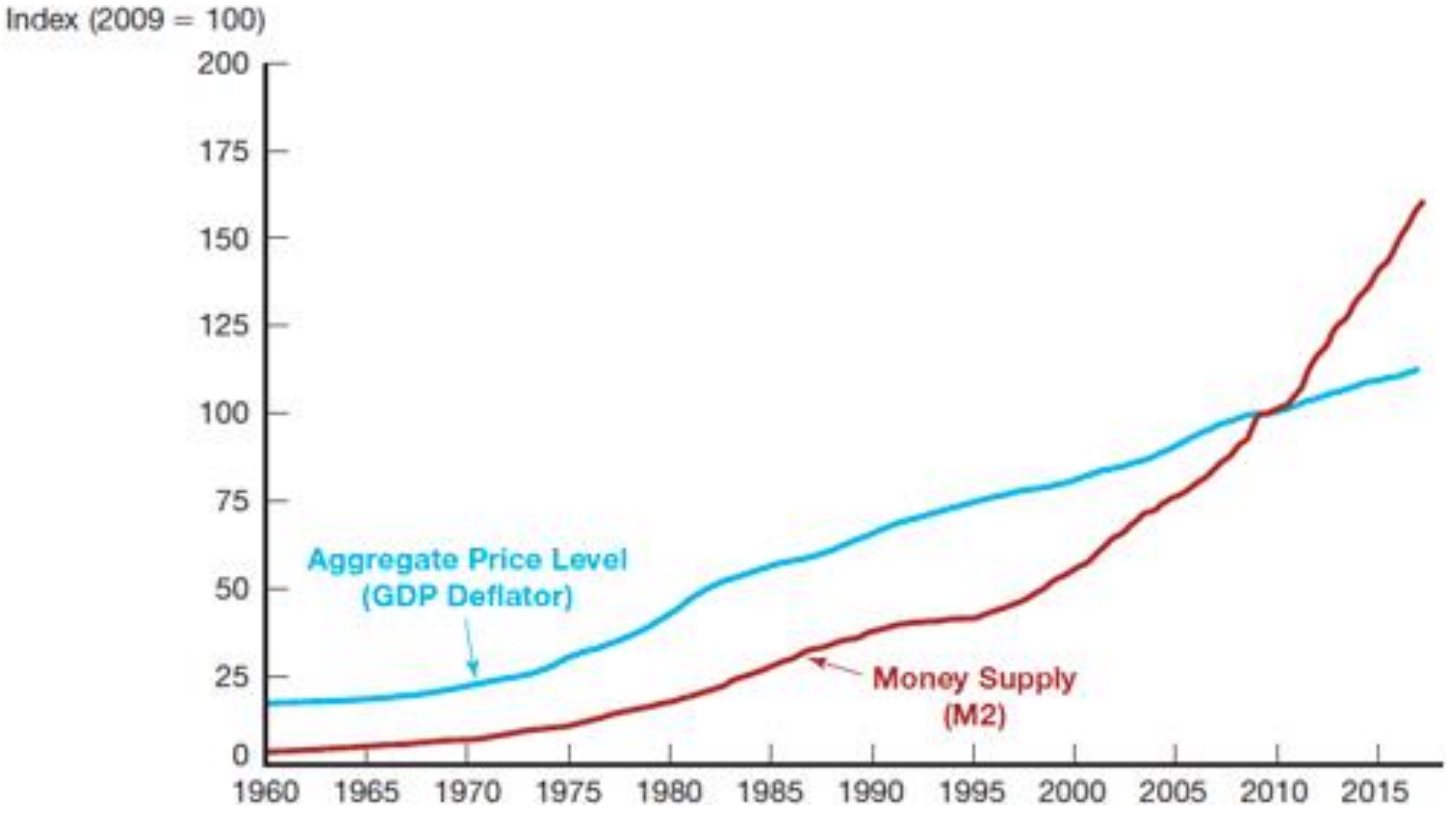
In Argentina, investment/saving is low, few people accept cash as payment when selling houses. Why?

Figure 3 Money Growth (M2 Annual Rate) and the Business Cycle in the United States, 1950–2017



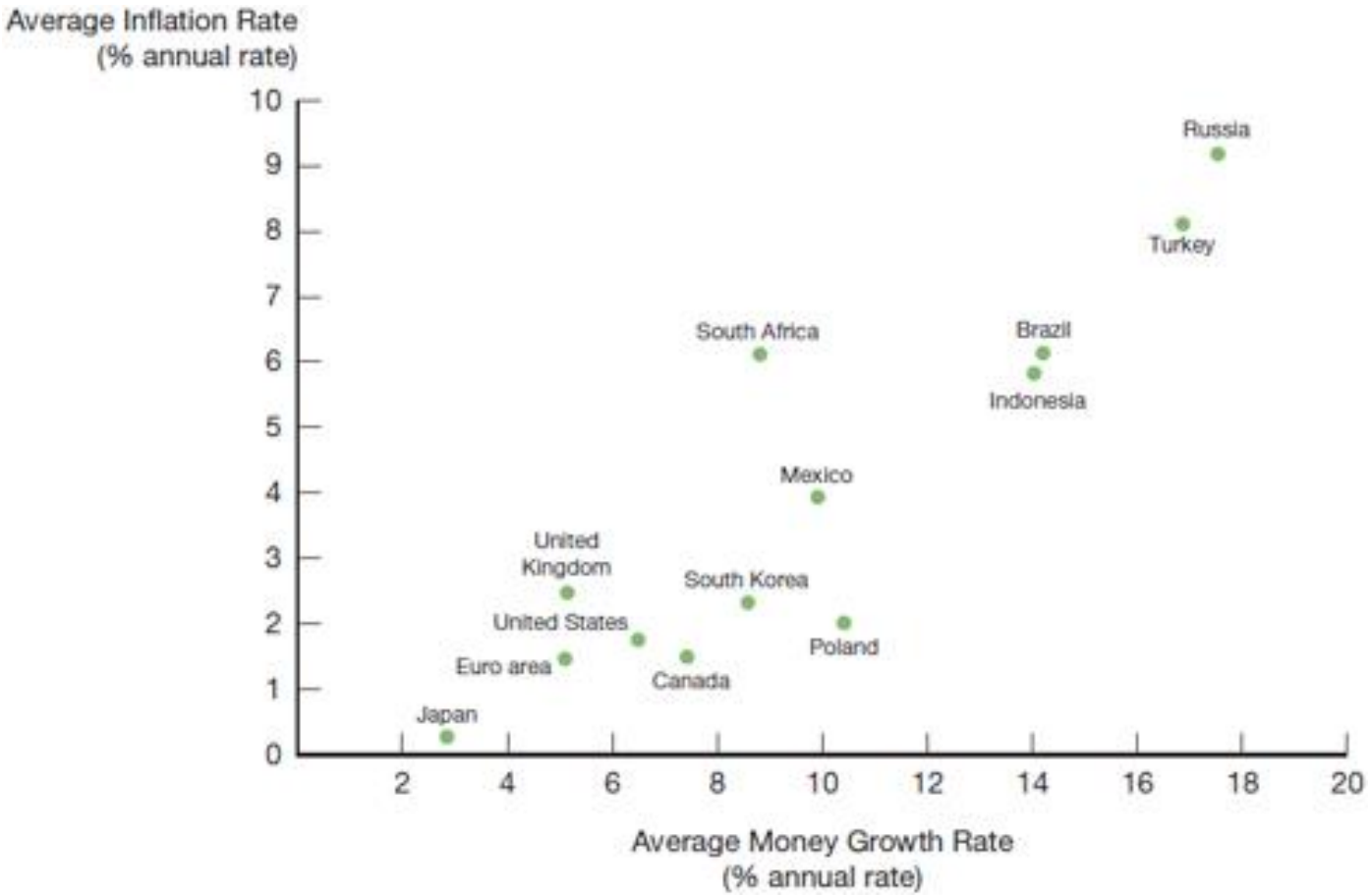
Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/M2SL

Figure 4 Aggregate Price Level and the Money Supply in the United States, 1960–2017



Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/M2SL; https://fred.stlouisfed.org/series/GDPDEF

Figure 5 Average Inflation Rate Versus Average Rate of Money Growth for Selected Countries, 2006–2016

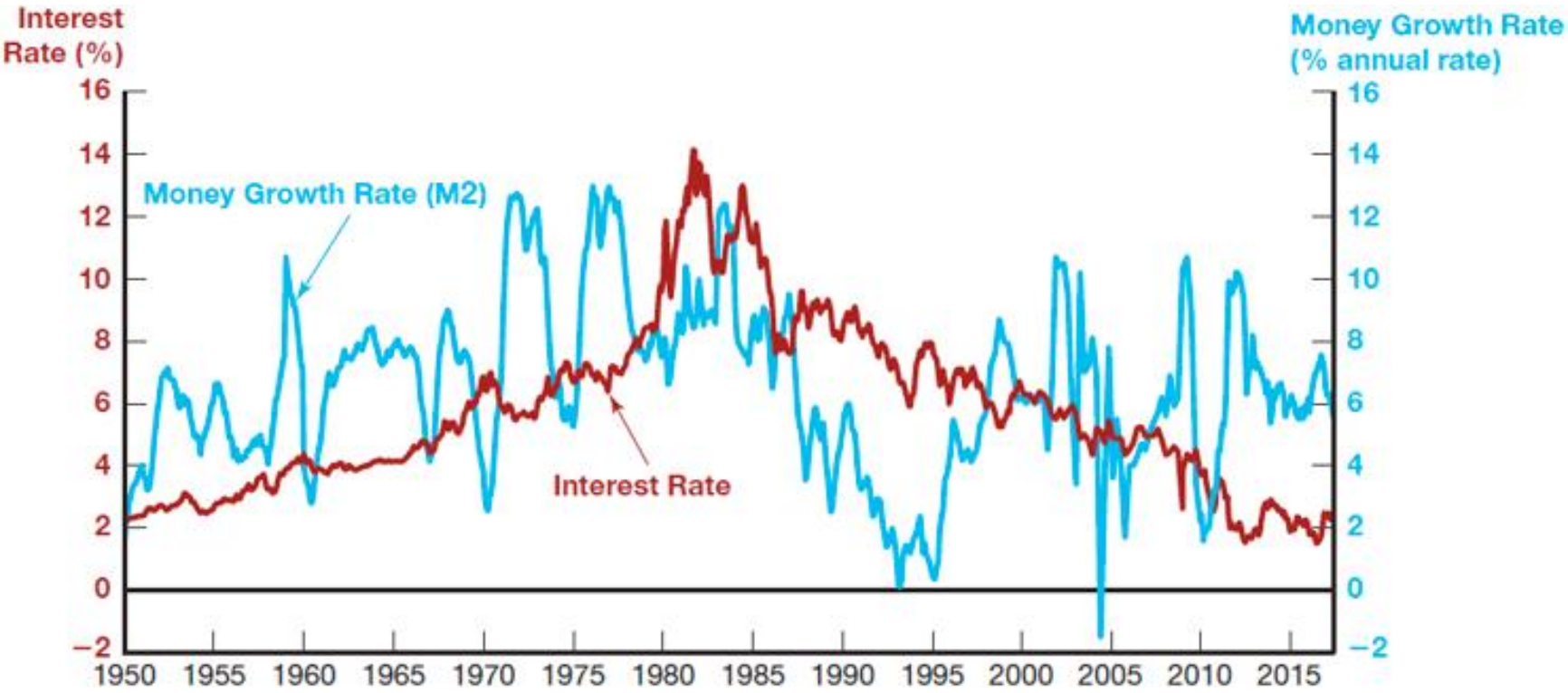


Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/

Money and Interest Rates

- Interest rates are the price of money
- Prior to 1980, the rate of money growth and the interest rate on long-term Treasury bonds were closely tied
- Since then, the relationship is less clear, but the rate of money growth is still an important determinant of interest rates

Figure 6 Money Growth (M2 Annual Rate) and Interest Rates (Long-Term U.S. Treasury Bonds), 1950–2017

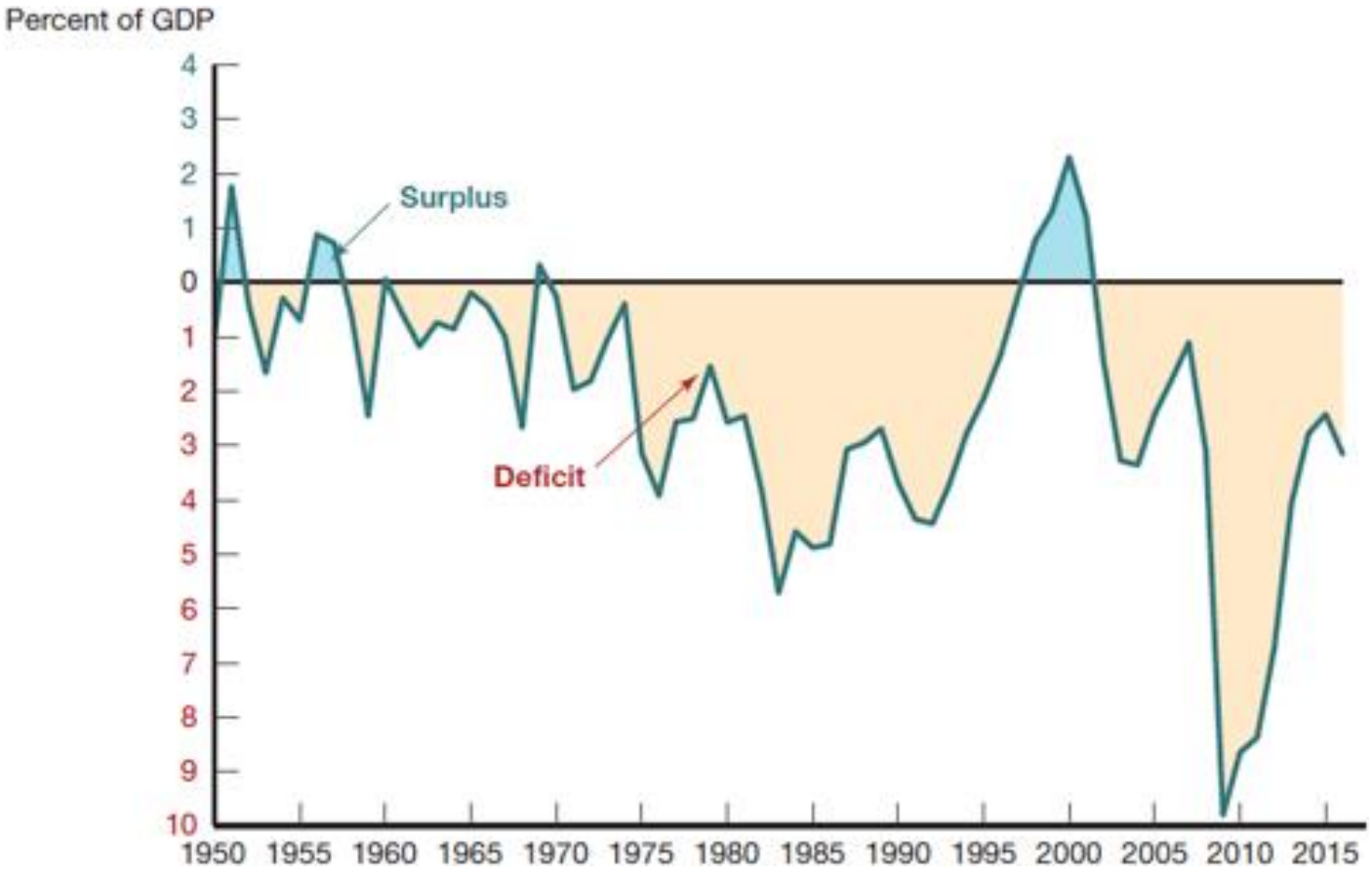


Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/M2SL; https://fred.stlouisfed.org/series/GS10; https://fred.stlouisfed.org/series/M2SL

Fiscal Policy and Monetary Policy

- **Monetary policy** is the management of the money supply and interest rates
  - Conducted in the United States by the Federal Reserve System (Fed)
- **Fiscal policy** deals with government spending and taxation
  - Budget deficit is the excess of expenditures over revenues for a particular year
  - Budget surplus is the excess of revenues over expenditures for a particular year
  - Any deficit must be financed by borrowing

Figure 7 Government Budget Surplus or Deficit as a Percentage of Gross Domestic Product, 1950–2016



Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/M2SL; https://fred.stlouisfed.org/series/FYFSGDA188Sl

The Foreign Exchange Market

- The **foreign exchange market**: where funds are converted from one currency into another
- The **foreign exchange rate** is the price of one currency in terms of another currency.
- The foreign exchange market determines the foreign exchange rate.

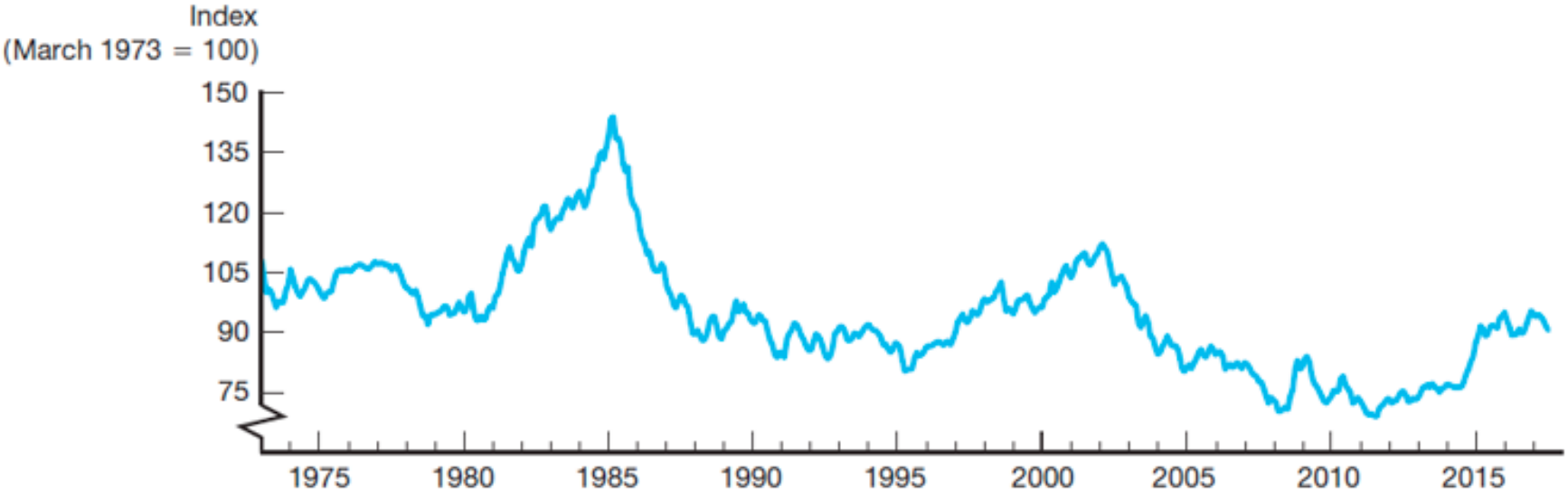
e.g., RMB versus USD

https://www.bloomberg.com/quote/USDCNY:CUR

Why Study International Finance

- Financial markets have become increasingly integrated throughout the world.
- The international financial system has tremendous impact on domestic economies:
  - How a country's choice of exchange rate policy affect its monetary policy?
  - How capital controls impact domestic financial systems and therefore the performance of the economy?
  - Which should be the role of international financial institutions like the IMF?

Figure 8 Exchange Rate of the U.S. Dollar, 1973–2017



Source: **Federal Reserve Bank of St. Louis, FRED database:** https://fred.stlouisfed.org/series/TWEXMMTH

The International Financial System

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